Analysis of Designed Experiments
STAT 5200/6200 (001), Fall 2026

Instructor Information

Jesse Wheeler
Mathematics and Statistics Department
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Office: Ray B. West (RWST) 421
Office Hours:

- Tuesday 10:30 AM – 11:45 AM.
- Thursday 1:00 PM – 3:00 PM.

You can contact me during office hours or via Email. I will try my best to respond to questions within 24 hours during Mon-Fri. Feel free to email me over the weekends, but I may not be actively monitoring my emails during these times and you may have longer response times.

Graduate Teaching Assistant

Samitha Herath
Statistics Ph.D. Student
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Office: FL L201 (In the link between RWST and Family Life)
Office Hours:

- Mondays 12:30 PM – 1:30 PM.
- Wednesday 2:00 PM – 3:00 PM.

Course Description

This course teaches analysis of balanced experimental designs with fixed, random, crossed, and nested factors; factorial, nested, nested factorial, split plot, split block, and repeated measures designs; and fixed and mixed effects models, residual analysis, and post hoc mean comparisons. Additional coursework is required for those enrolled in the graduate-level course.

PREREQ: Earn a minimum grade of C- one of the following:

- STAT 2000, STAT 3000, DATA 3100, PSY 3010, SOC 3120

This course is 3 Credit Hours, and we meet every Tuesday and Thursday, 9:00 am - 10:15 am, Engineering 106, with the exception of university-observed holidays and breaks.

Course Objectives and Outcomes

This class will teach the design of experiments, and the analysis of real data sets that arise. Datasets are provided from a wide range of disciplines. At the end of the class, you will be able to:

- Recognize most of the commonly used experimental designs, including:
  - Completely Randomized Designs.
  - Factorial Designs.
  - Block Designs.
  - Split-plot and split-block designs.
  - Repeated Measures designs.
  - Nested Designs.
- You will be able to identify situations in which each of these kinds of designs is applicable.
- You will be able to construct and analyze each of these kinds of design.

Course Format

This is a standard class where we will meet in-person. All students are required to attend all classes during the first week. Any student having to miss during the first week needs instructor approval. Students will be dropped from their courses if they do not attend the first week and do not have permission from the instructor.

After the first week of classes, attendance is expected but will not be recorded. Lectures will not be recorded, but slides will be available to students who miss lecture. Course materials, activities, and assignments will be posted on Canvas.

Canvas

USU Canvas will be the primary tool for receiving announcements, assignments, and course lectures. Homework assignments will be uploaded to Canvas, along with an Canvas announcement that a new homework assignment has been posted.

The Canvas course will be broken down into separate modules, each representing a core topic for the course. In each module, there will be three basic resources for students:

- Lecture Slides
- Accompanying lecture notes
- Annotated Slides

The lecture slides will be made available prior to the lecture. I will try my best to get them posted on Canvas the day before lectures, but as it is my first time teaching this course and I have to make the material from scratch, I may have some last minute changes the day of lectures. After our lectures, I will upload a version of the slides–with my own notes and comments–to a file called "annotated slides".

In addition to the lecture slides, there are accompanying lecture notes, which contain the same material as the slides, plus a little extra (e.g., solutions to certain in class problems).

Course Website

In addition to Canvas, course material will be available on my own personal course website. You may find this is easier to navigate course material, though using this website is not required. Access to the website will remain indefinitely after the course has concluded, whereas Canvas material may be lost once you graduate.

General course information can be found at the course website: `https://jeswheel.github.io/5200_f26`

Assessment Methods

Student understanding will be assessed via 8 homework assignments, 1 midterm exam, and a final exam. In addition to formal assessment methods, students will also be graded based on course participation. A full breakdown of assessment category weighting is given in Table 1.

Homework

The homework assignments may not all be worth the same number of points, nor be the same length. They will be given (approximately) every other week. Late homework submissions will not be accepted, unless permission is provided by the instructor.

Grade	Percentage Range
Participation	10%
Homework	35%
Midterm 1	25%
Final	30%

Table 1: Assignment Weights

Homeworks will involve a variable number of points, depending on the assignment. Each homework assignment will have a single point for "*academic integrity*". This is simply a statement about what resources you used, how you benefited from them, and how the final results are still your own.

You can use any resources to help you with your homework, including the internet and generative AI. Before using outside sources, you are required to work on a given problem for at least one hour (honor-system).

The order of preferred sources are:

- Lecture Slides, Lecture Notes, and the Textbook.
- Office hours.
- Classmates.
- The internet (statsoverflow).
- AI.

When looking for outside sources, please try your best to not search directly for a solution. To obtain full points for "academic integrity", each homework assignment must include a statement at the start of the assignment indicating which sources they used, how long they worked on the problem before looking for alternative sources, and what they learned from the source they found. More details regarding the 1 academic integrity point can be found at `https://jeswheel.github.io/5200_f26/rubric_homework.html`

Exams

- **Midterm 1**: Planned for **Oct 29**, during regular class time.
- **Final Exam**: Thursday, Dec 17, 9:30 a.m. – 11:20 a.m. The location will be our regular classroom, and the exam will be cummulative.

Please see USU Final Exam Schedule Webpage for more details regarding final exam scheduling.

Participation

Now more than ever, active engagement with course material is a critical part of a University experience. This class should have a relatively low workload for a 5000 / 6000 level class, but I do expect active participation. There will be four participation "assignments" on Canvas, each based on completion. These assignments will be broken into 4 different periods:

- Sep 03 – Sep 25
- Sep 26 – Oct 23
- Oct 24 – Nov 12
- Nov 13 – Dec 11

These should be free points, but is designed to encourage real-engagement with the course material. Examples of how you can earn participation points include: Attending office hours, asking questions about the reading material, answer questions another student has, a brief write-up on how you have used what you learned outside the classroom, an email to me (questions or comments) regarding the weekly reading etc. Feel free to be creative!

Textbook and Course Materials

Textbook reading is expected, though not directly graded. Comments or questions regarding reading assigments can be used to satisfy *participation* points. We will use the following textbook in this class:

Oehlert, G. W., 2000. *A First Course in Design and Analysis of Experiments.* W. H. Freeman. ISBN: 0-7167-3510-5. .

This text is freely available online under a "Creative Commons" license: `http://users.stat.umn.edu/~gary/book/FCDAEdraft2021.pdf` Briefly, you are free to copy, distribute, and transmit the text, provided thta you properly attribute the work, do not use it for commercial purposes, and do not alter, transform, or build upon the work.

Software

For Stat 5200, you will likely need to use some statistical software to complete your homework assignments. I will provide example code and demonstrations using the R programming language; for the purpose of our class, think of this software as a special calculator used for Statistics. To interact with this software, I highly recommend using RStudio, an Integrated Development Environment (IDE) built for writing R code.

Why R? R and RStudio are both free, and widely used software in both academia and industry. R has been built specifically to help practicioners do statistics easily.

- Installing R: https://cran.r-project.org/
- Installing RStudio: https://posit.co/download/rstudio-desktop/. Note that you should install R prior to installing RStudio.

Other types of calculators or programming languages are permitted, but it is not anticipated that they will be beneficial beyond what you can do in R.

Grading Scale

The grading scale follows the USU standard scale, in Table 2.

Grade	Percentage Range
A	93–100
A-	90–92.99
B+	87–89.99
B	83–86.99
B-	80–82.99
C+	77–79.99
C	73–76.99
C-	70–72.99
D+	67–69.99
D	60–66.99
F	0–59.99

Table 2: Grade Breakdown

Course Topics (Tentative)

We will closely follow the textbook, but not necessarily cover every chapter or in the same order. Our tentative plan is

- **Module 1**: Introduction and Review (Chapter 1 of Oehlert, G. W. (2020)).
- **Module 2**: Randomization and Hypothesis Testing (Chapter 2 of Oehlert, G. W. (2020)).
- **Module 3**: Completely Randomized Designs and ANOVA (Chapter 3 of Oehlert, G. W. (2020)).
- **Module 4**: Factorial Designs (Chapter 8 of Oehlert, G. W. (2020)).
- **Module 5**: Random Effects Models (Chapter 11 of Oehlert, G. W. (2020)).
- **Module 6**: General Mixed Effects Models (Chapter 12 of Oehlert, G. W. (2020)).

- **Module 7**: Designs with Nested Factors (Chapter 12 of Oehlert, G. W. (2020)).
- **Module 8**: Block Designs (Chapter 13 of Oehlert, G. W. (2020)

Schedule (Tentative)

The table below provides a tentative schedule of the course. This table will be updated on the course website as needed.

Week	**Dates (T Th)**	**Notes**
1	Sep 1, 3	
2	Sep 8, 10	
3	Sep 15, 17	
4	Sep 22, **24***	*Participation Due
5	Sep 29, Oct 1	
6	Oct 6, 8	
7	Oct 13, 15	
8	Oct 20, **22***	*Participation due
9	Oct 27, **29**	**Midterm 1**
10	Nov 3, 5	
11	Nov 10, **12***	*Participation due
12	Nov 17, 19	
13	Nov 24, 26	**Thanksgiving - No Class**
14	Dec 1, 3	
15	Dec 8, **10***	No-Test Week, *Participation Due
16	Dec 14–18	Finals Week (**Dec 17**)

For other important dates and deadlines, please see the University academic calendar: `https://www.usu.edu/calendar/academic/?year=2026`.

Academic Freedom and Professional Responsibilities

Academic freedom is the right to teach, study, discuss, investigate, discover, create, and publish freely. Academic freedom protects the rights of faculty members in teaching and of students in learning. Freedom in research is fundamental to the advancement of truth. Faculty members are entitled to full freedom in teaching, research, and creative activities, subject to the limitations imposed by professional responsibility. USU Policy 4002 further defines academic freedom and professional responsibilities.

Academic Integrity

Each student has the right and duty to pursue his or her academic experience free of dishonesty. To enhance the learning environment at Utah State University and to develop student academic integrity, each student agrees to the following Honor Pledge:

> "I pledge, on my honor, to conduct myself with the foremost level of academic integrity."

A student who lives by the Honor Pledge is a student who does more than not cheat, falsify, or plagiarize. A student who lives by the Honor Pledge:

- Espouses academic integrity as an underlying and essential principle of the Utah State University community;
- Understands that each act of academic dishonesty devalues every degree that is awarded by this institution; and
- Is a welcomed and valued member of Utah State University.

Academic Dishonesty

The instructor of this course will take appropriate actions in response to Academic Dishonesty, as defined the University's Student Code. Acts of academic dishonesty include but are not limited to:

- **Cheating:** using, attempting to use, or providing others with any unauthorized assistance in taking quizzes, tests, examinations, or in any other academic exercise or activity. Unauthorized assistance includes:
  - Working in a group when the instructor has designated that the quiz, test, examination, or any other academic exercise or activity be done "individually;"
  - Depending on the aid of sources beyond those authorized by the instructor in writing papers, preparing reports, solving problems, or carrying out other assignments;
  - Substituting for another student, or permitting another student to substitute for oneself, in taking an examination or preparing academic work;
  - Acquiring tests or other academic material belonging to a faculty member, staff member, or another student without express permission;
  - Continuing to write after time has been called on a quiz, test, examination, or any other academic exercise or activity;
  - Submitting substantially the same work for credit in more than one class, except with prior approval of the instructor; or engaging in any form of research fraud.
- **Falsification:** altering or fabricating any information or citation in an academic exercise or activity.

- **Plagiarism:** representing, by paraphrase or direct quotation, the published or unpublished work of another person as one‘s own in any academic exercise or activity without full and clear acknowledgment. It also includes using materials prepared by another person or by an agency engaged in the sale of term papers or other academic materials.

For this class, the use of AI and other outside resources is allowed for Homework, but not for exams. When using any outside resources for homework (including peers, the internet, or AI), you must:

- Be clear about how and why you used this resource.
- Be clear about how the work is still your own (not just copied from somewhere), and how you still engaged with the homework problem.

For more precise examples and details, please see the general course grading rubric: `https://jeswheel.github.io/5200_f26/rubric_homework.html`.

Withdrawal Policy and "I" Grade Policy

Students are required to complete all courses for which they are registered by the end of the semester. In some cases, a student may be unable to complete all of the coursework because of extenuating circumstances, but not due to poor performance or to retain financial aid. The term 'extenuating' circumstances includes: (1) incapacitating illness which prevents a student from attending classes for a minimum period of two weeks, (2) a death in the immediate family, (3) financial responsibilities requiring a student to alter a work schedule to secure employment, (4) change in work schedule as required by an employer, or (5) other emergencies deemed appropriate by the instructor.